

Structure of solutions of linear ODEs

1. Linear ODEs in state-space form

Consider the normalized n th-order equation

$$y^{(n)} + a_{n-1}(t)y^{(n-1)} + \cdots + a_1(t)\dot{y} + a_0(t)y = g(t). \quad (1)$$

Choose the state. Complete the state vector and its derivative.

$$x = \begin{pmatrix} \\ \\ \\ \end{pmatrix}, \quad \dot{x} = \begin{pmatrix} \\ \\ \\ \end{pmatrix}.$$

Identify $A(t)$ and $G(t)$. Fill in the last row and the forcing term.

$$\dot{x} = \underbrace{\begin{pmatrix} & & \cdots & \\ & & \cdots & \\ & & \cdots & \\ & & \cdots & \end{pmatrix}}_{A(t)} x + \underbrace{\begin{pmatrix} \\ \\ \\ \end{pmatrix}}_{G(t)}.$$

Example 10.1 – Write a concrete system. Rewrite $\ddot{y} + 3\dot{y} + 2y = \sin t$ as a first-order system. State how $y(t_0) = \eta_0$ and $\dot{y}(t_0) = \eta_1$ become an initial condition for x .

Check the regularity assumptions for $F(t, x) = A(t)x + G(t)$.

(a) F is _____ on $I \times \mathbb{R}^n$ if and only if _____.

(b) The partials _____ = _____ are _____ on $I \times \mathbb{R}^n$ if and only if _____.

Remark. In conclusion, _____ ensures applicability of the existence and uniqueness theorem to a Linear ODE $y^{(n)} + a_{n-1}(t)y^{(n-1)} + \cdots + a_1(t)\dot{y} + a_0(t)y = g(t)$.

Remark. The existence and uniqueness theorem applies to each initial condition $(t_0, x_0) \in$ _____. However, recall that it just tells us that there exists a unique solution **on some interval** (_____, _____).

2. Existence on the entire coefficient interval

Theorem 10.1 (Existence and uniqueness for linear ODEs). Suppose $a_0, \dots, a_{n-1}, g$ in (1) are continuous on an open interval $I = (a, b)$. For every $t_0 \in I$ and every $(\eta_0, \dots, \eta_{n-1}) \in \mathbb{R}^n$, the IVP with $y^{(j)}(t_0) = \eta_j$ for $j = 0, \dots, n-1$ has a _____ defined on _____.

How is this stronger than the general existence theorem?

Example 10.2. For $\dot{y} + (\sin t)y = \cos t$ with $y(t_0) = y_0$, identify a_0 , g , and the largest open interval on which the theorem guarantees a unique solution. Do not solve the equation.

Example 10.3. Consider $(t-1)\dot{y} + y = \sin t$ with $y(t_0) = y_0$. identify the largest open interval on which the theorem guarantees a unique solution.

(a) Normalize the equation. Where are its coefficients continuous?

(b) State the theorem's conclusion separately for $t_0 < 1$ and $t_0 > 1$.

Takeaway: what must you do before applying the linear theorem?

4. Why is the dimension equal to the order?

Fix $t_0 \in I$. By existence and uniqueness, a solution is uniquely determined by its initial values at t_0 . Define the map

$$\Phi : S \longrightarrow \mathbb{R}^n, \quad \Phi(y) = \begin{pmatrix} y(t_0) \\ \dot{y}(t_0) \\ \vdots \\ y^{(n-1)}(t_0) \end{pmatrix}.$$

Show that Φ is an isomorphism.

(a) Linearity. Complete the identity and justify it.

For $y_1, y_2 \in S$ and $c_1, c_2 \in \mathbb{R}$,

$$\Phi(c_1 y_1 + c_2 y_2) = \underline{\hspace{10cm}}.$$

(b) Surjectivity. Let $Y \in \mathbb{R}^n$ be arbitrary.

Why does there exist a solution $y \in S$ such that $\Phi(y) = Y$? Which part of the existence and uniqueness theorem is used?

(c) Injectivity. Suppose $\Phi(y_1) = \Phi(y_2)$.

Why must $y_1 = y_2$ on I ? Which part of the theorem is used?

Conclude the proof.

Since Φ is linear, onto, and one-to-one, it is an . Consequently,

$$\dim S = \dim \mathbb{R}^n = \underline{\hspace{2cm}}.$$